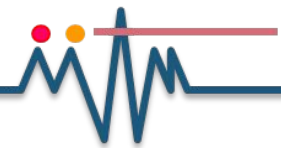


Real Time Experimental Extended Range Prediction

BASED ON

INITIAL CONDITION: 17th June 2026

**Working group on Extended Range Prediction
Ministry of Earth Sciences, Govt. of India**



Disclaimer

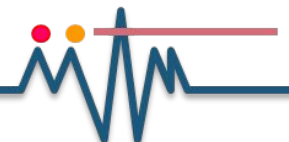
This PPT is based on the forecasts from the second generation extended range prediction system (ERPv2) developed at IITM with a multiphysics strategy (Sahai et al. 2021; Kaur et al. 2022; Dey et al. 2025). The forecasts are being run on an experimental basis for the monsoon season 2022.

The materials contained in this PPT are made from raw model output and are research products. They are meant for scientific use only.

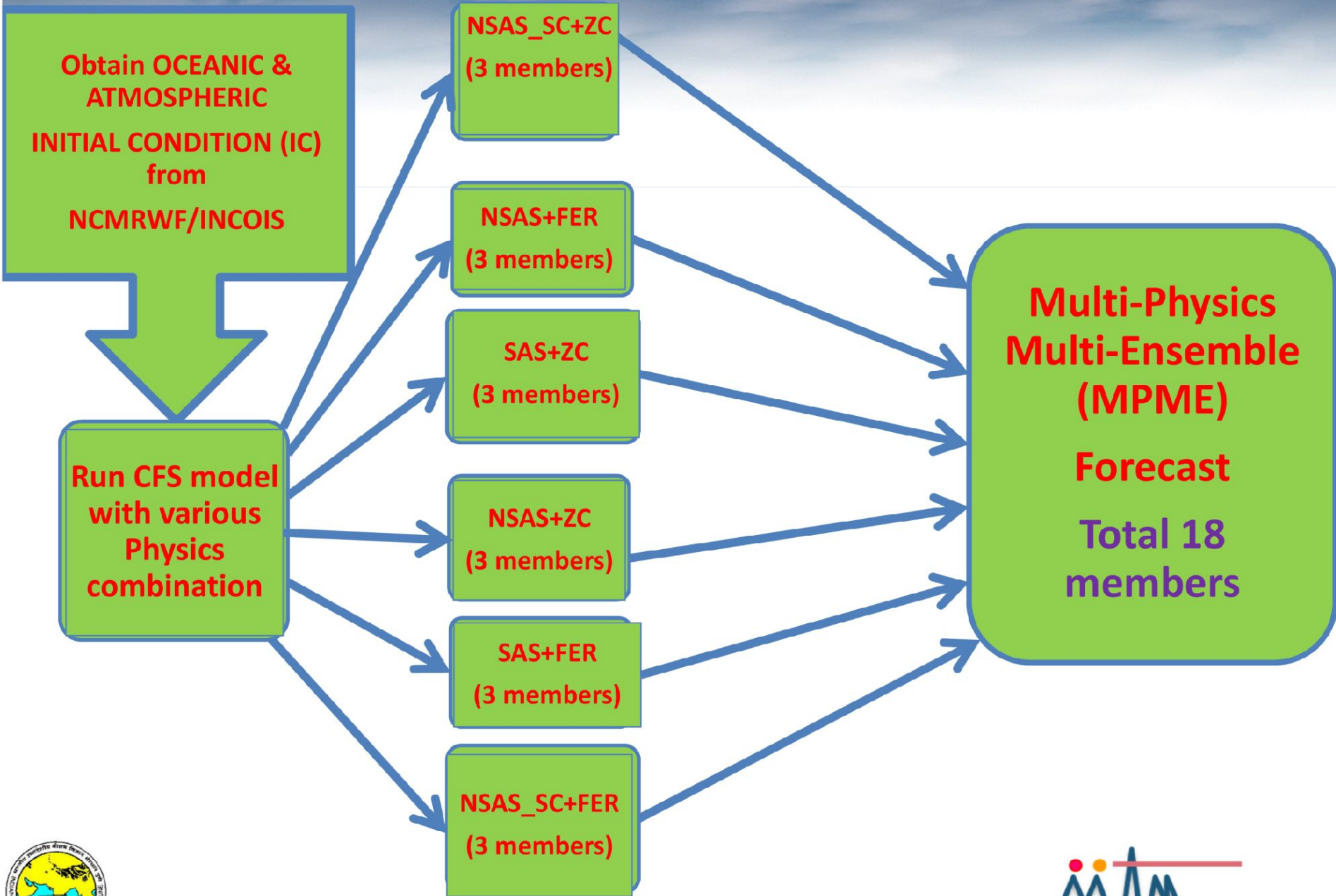
For forecasts based on ERPv1, developed by IITM and being run operationally by IMD, please visit,

https://nwp.imd.gov.in/cfs_rf.php

https://mausam.imd.gov.in/imd_latest/contents/extendedrangeforecast.php



ERPv2 (Experimental setup)



Multi-physics framework

Convection

Microphysics

Seamless Horizontal Resolution

**Simplified
Arakawa
Shubert
(SAS)**

**Revised
Simplified
Arakawa
Shubert
(NSAS)**

**Revised
Simplified
Arakawa
Shubert
Shallow
convection**

**Zhao &
Car
(ZC)**

**Ferrier
(FER)**

T574

T382

HIGH

LOW

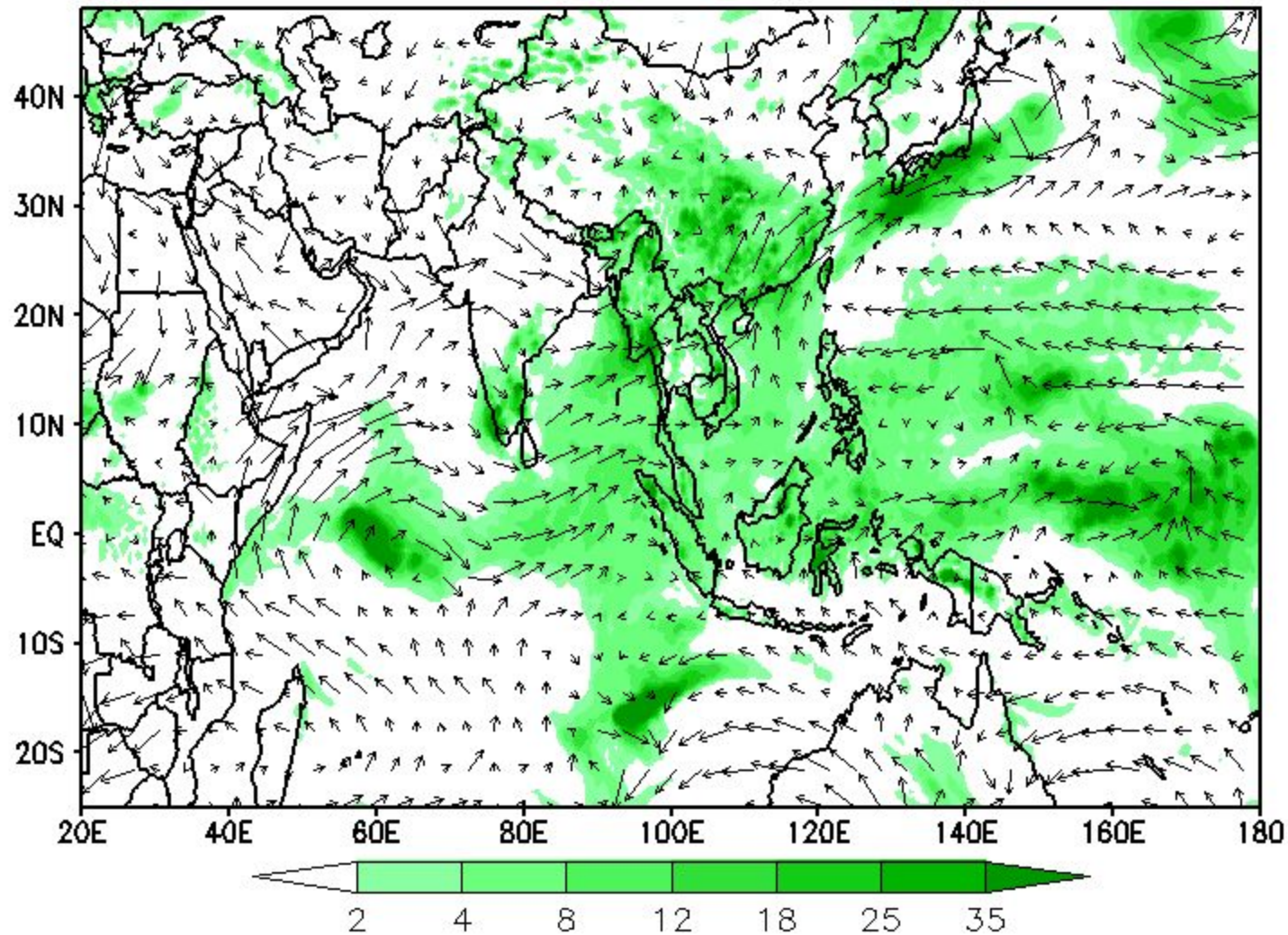


Daily evolution of rainfall and wind at 850hPa (by MPME)

MPME Forecast Valid Time = 00Z18JUN2026

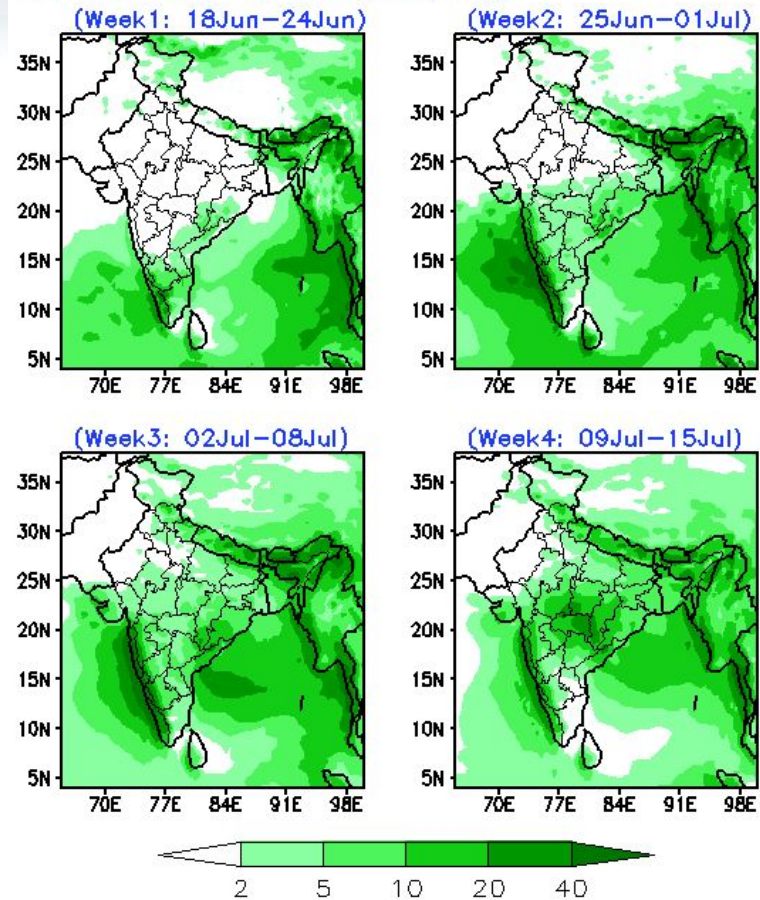
Initial Condition : 20260617

Rainfall (shaded, mm/day) & 850hPa winds (vector, 20)

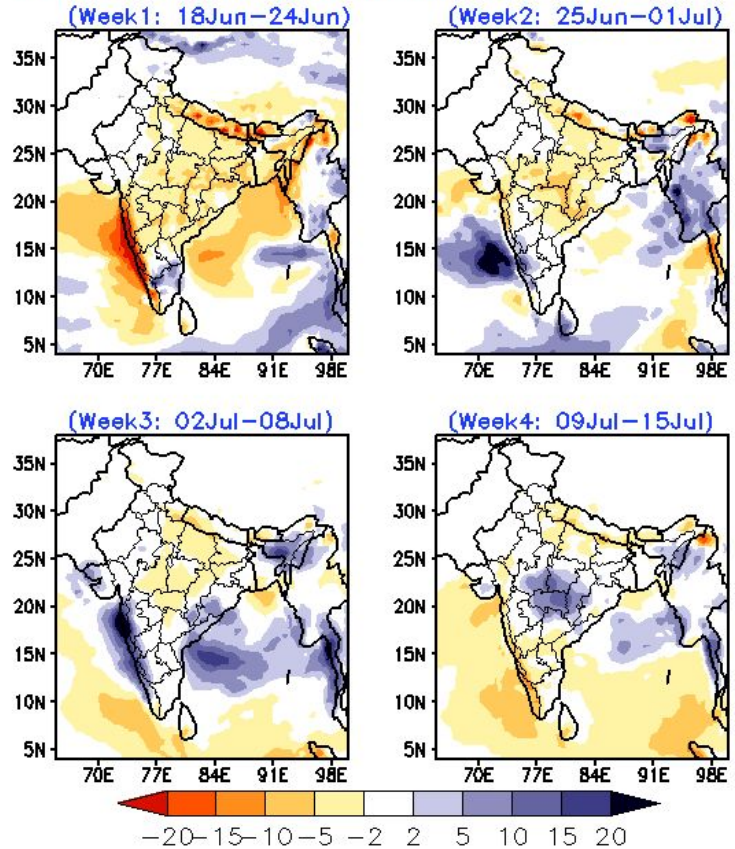


Predicted week wise rainfall (by MPME)

MPME Actual Rainfall (mm/day) IC=20260617



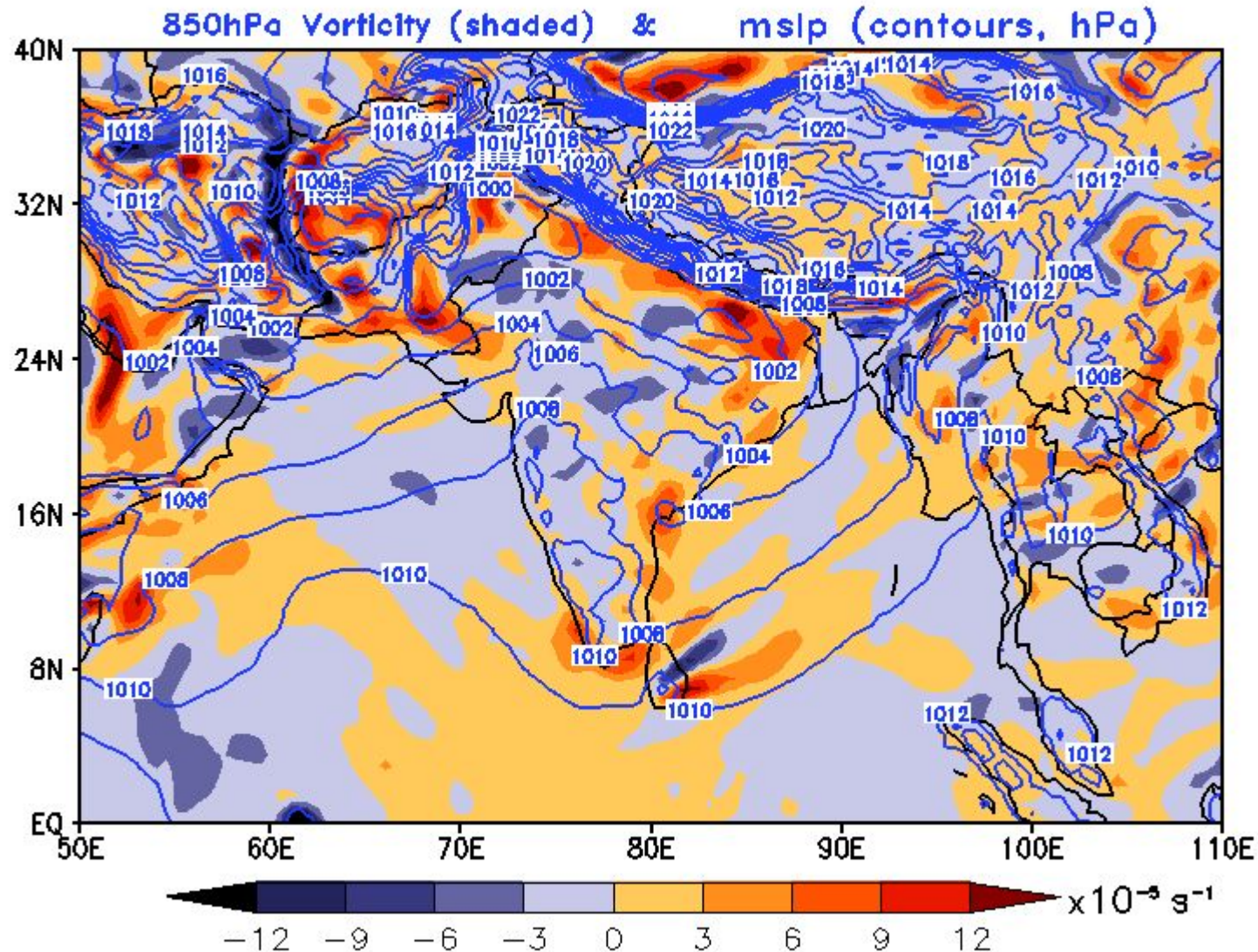
MPME Rainfall Anomaly (mm/day) IC=20260617



Daily evolution of vorticity at 850hPa and mean sea level pressure (by MPME)

MPME, Forecast Valid Time = 00Z18JUN2026

Initial Condition : 20260617

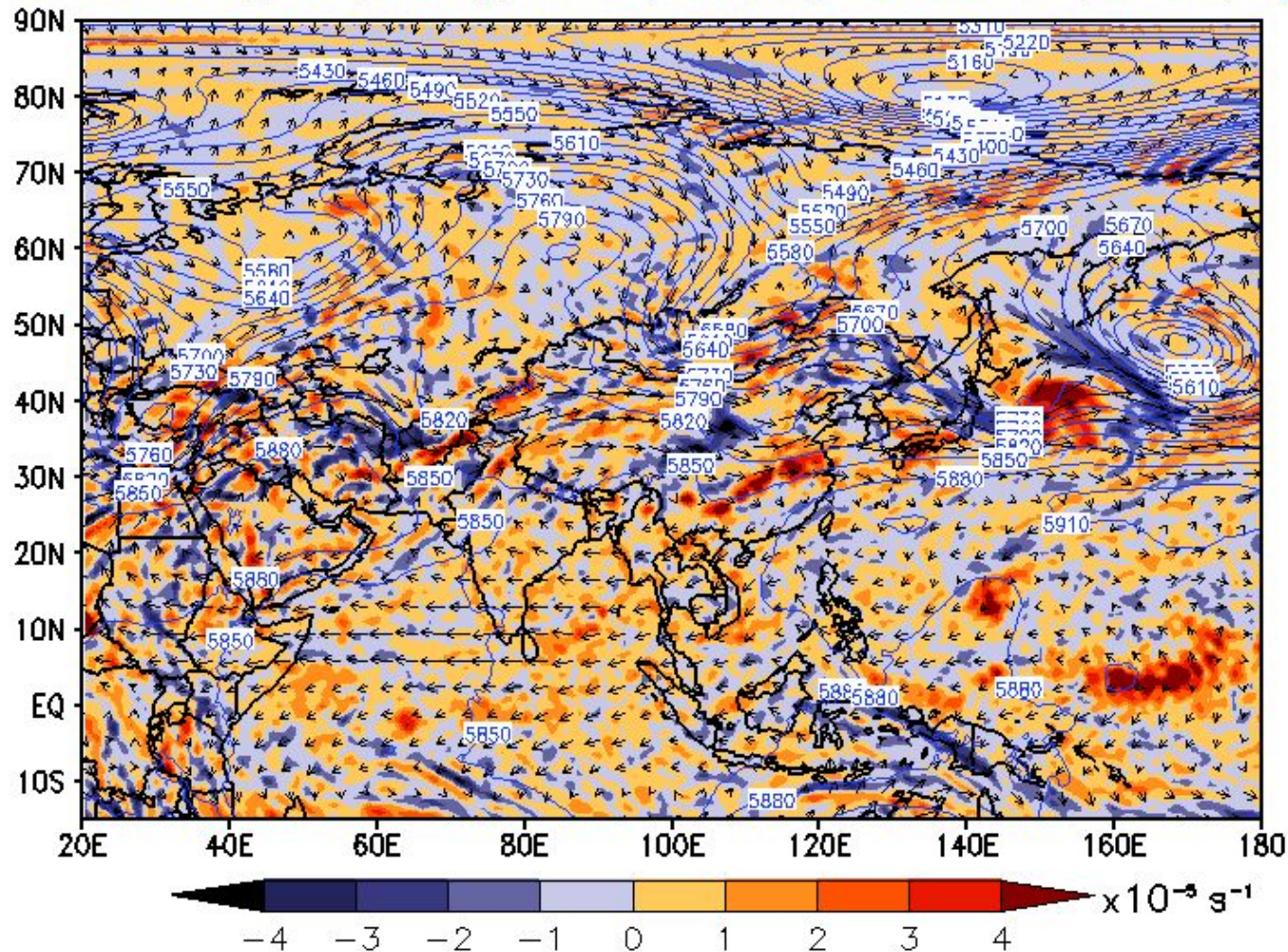


Daily evolution of divergence and winds at 200hPa and 500mb Geopotential Height (by MPME)

MPME, Forecast Valid Time = 00Z19JUN2026

Initial Condition : 20260617

200 hPa Divergence (shaded), Winds (vector, 50) & 500mb GH (contours, m)

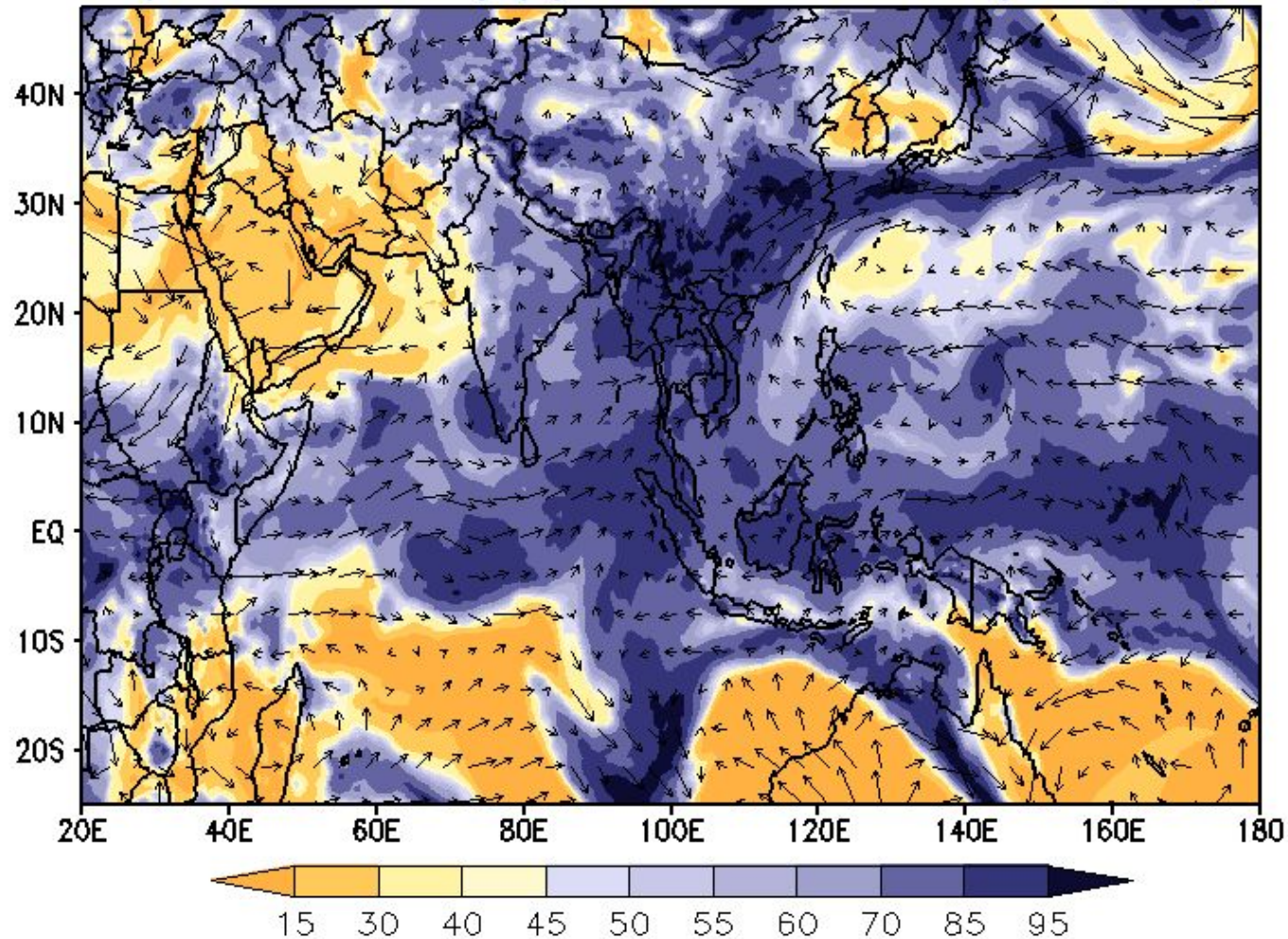


Daily evolution of RH and winds at 700hPa (by MPME)

MPME Forecast Valid Time = 00Z19JUN2026

Initial Condition : 20260617

700hpa Relative humidity (%) & 700hPa winds (vector, 20°)

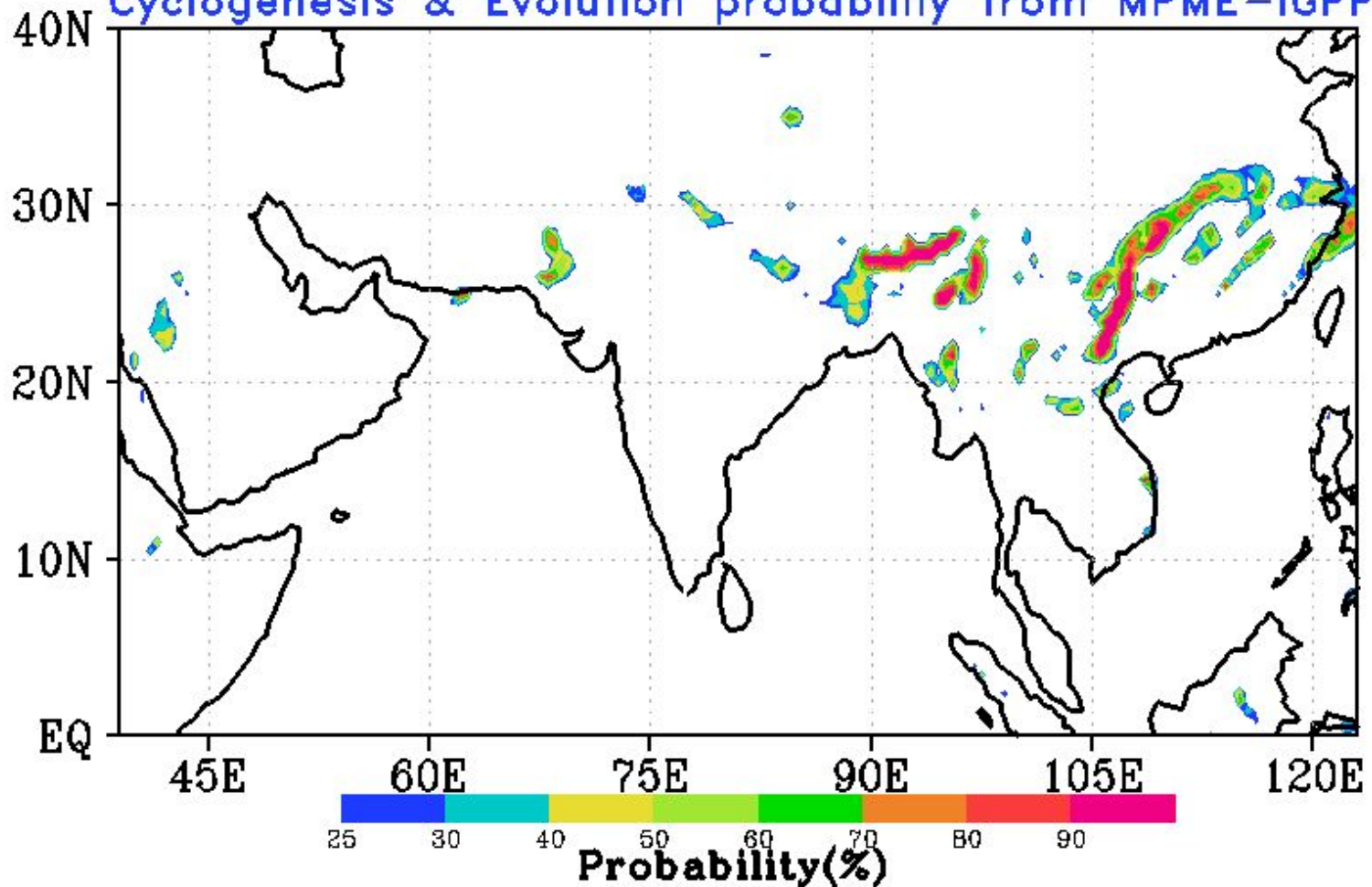


Forecast of Cyclogenesis probability based on GPI from MPME

MPME, Forecast Valid Time = 00Z18JUN2026

Initial Condition : 20260617

Cyclogenesis & Evolution probability from MPME-IGPP

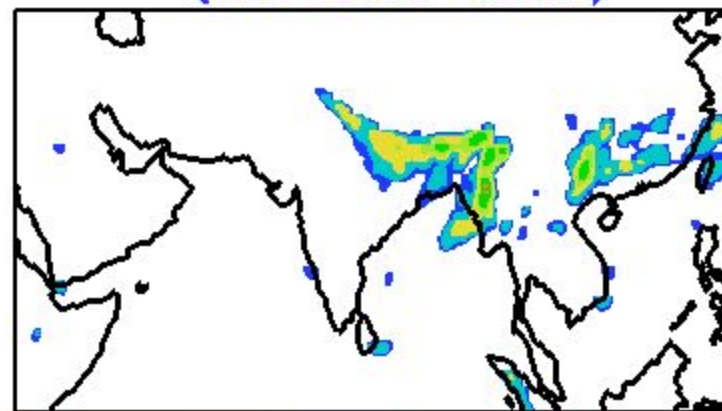
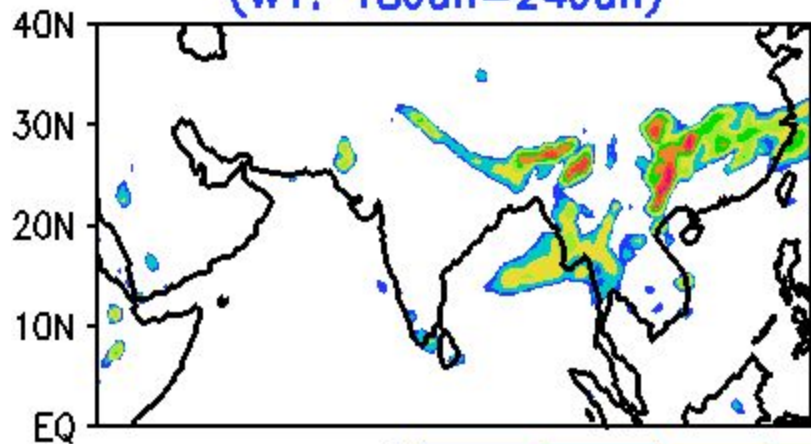


Cyclogenesis & Evolution Probability (%) from MPME-IGPP

Initial Condition : 20260617

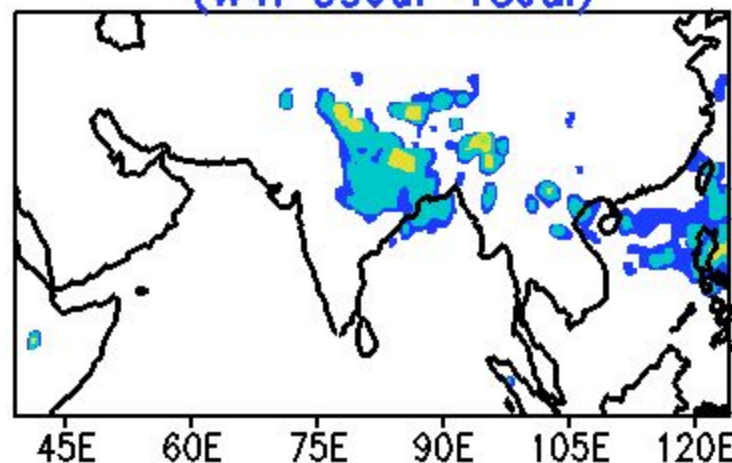
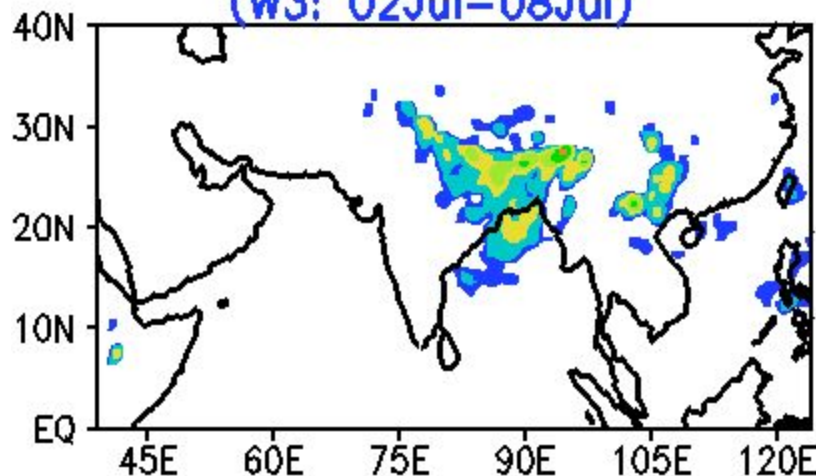
(W1: 18Jun-24Jun)

(W2: 25Jun-01Jul)



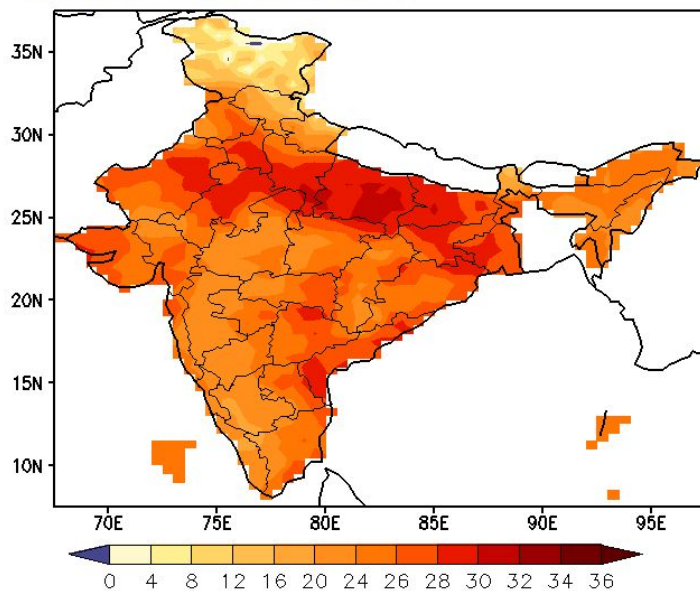
(W3: 02Jul-08Jul)

(W4: 09Jul-15Jul)

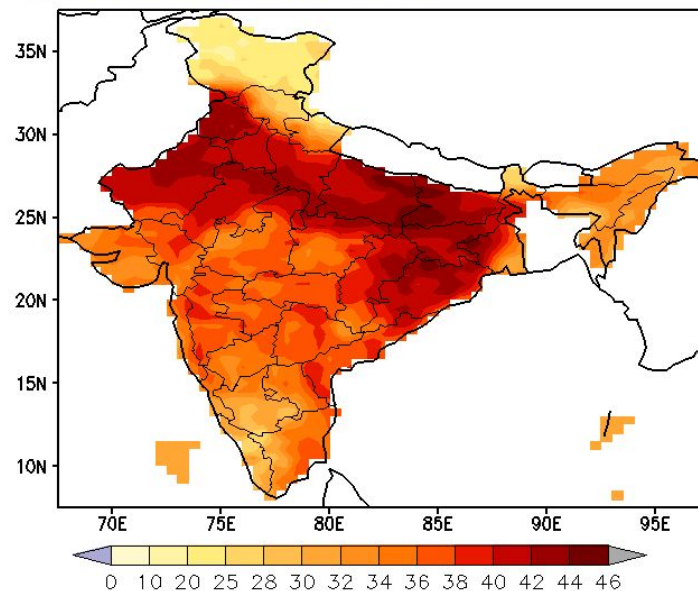


Daily evolution of minimum and maximum temperature (by MME)

MPME, Forecast Valid Time = 00Z18JUN2026
Minimum Temp. Actual in °C (IC=20260617)



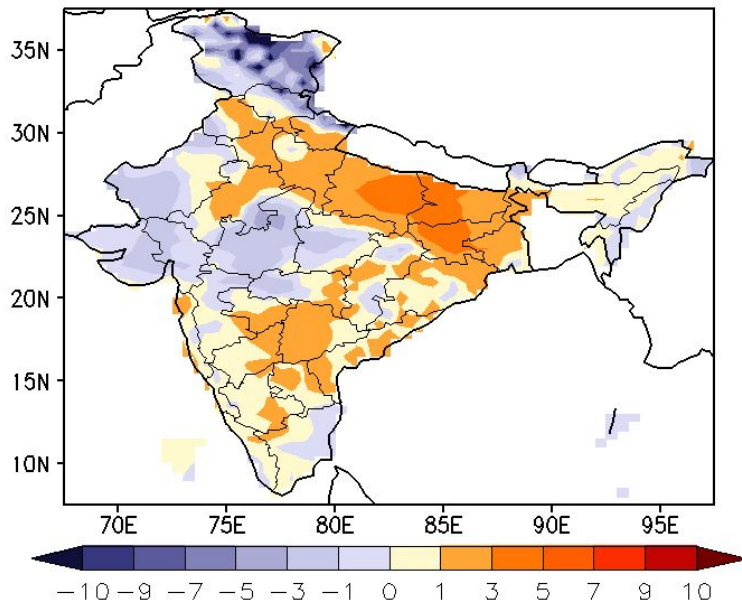
MPME, Forecast Valid Time = 00Z18JUN2026
Maximum Temp. Actual in °C (IC=20260617)



Daily evolution of minimum and maximum temperature Anomaly (by MME)

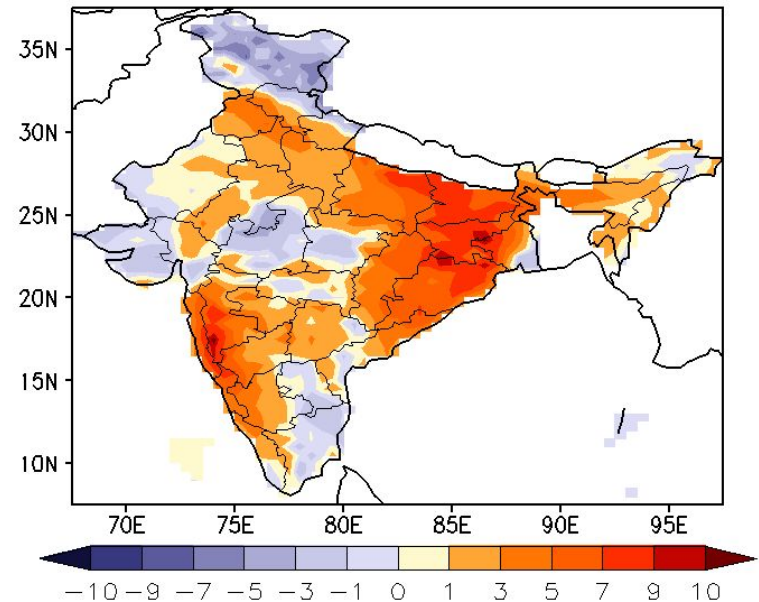
MPME,Forecasted Min. Temp Anomaly,Valid Time=00Z18JUN2026

Minimum Temp. Anomaly in °C (IC=20260617)



MPME,Forecasted Max. Temp Anomaly,Valid Time=00Z18JUN2026

Maximum Temp. Anomaly in °C (IC=20260617)

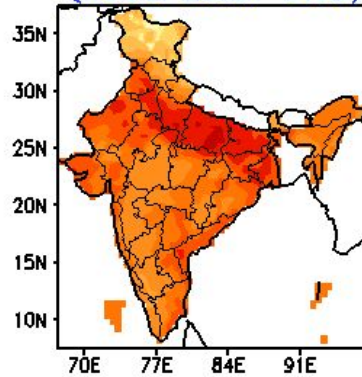
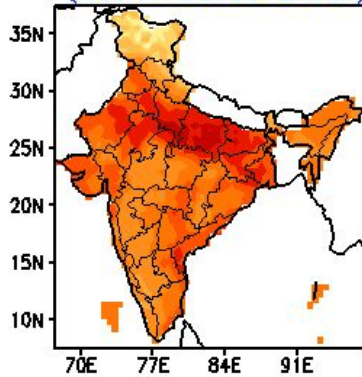


Predicted week wise temperature actual (by MPME)

Minimum Temperature Actual (°C) IC=20260617

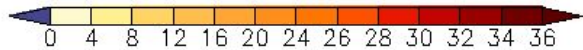
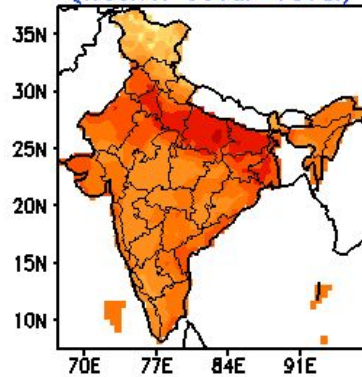
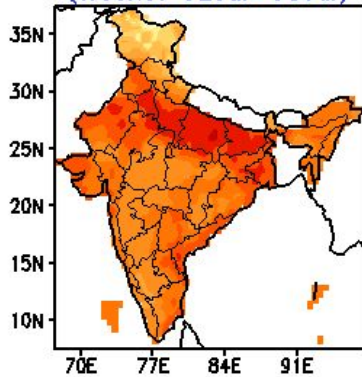
(Week1: 18Jun-24Jun)

(Week2: 25Jun-01Jul)



(Week3: 02Jul-08Jul)

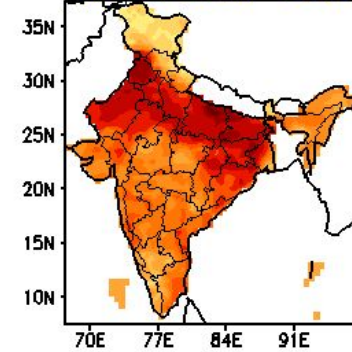
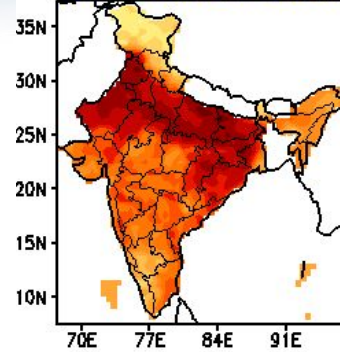
(Week4: 09Jul-15Jul)



Maximum Temperature Actual (°C) IC=20260617

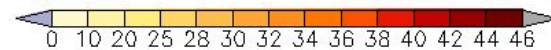
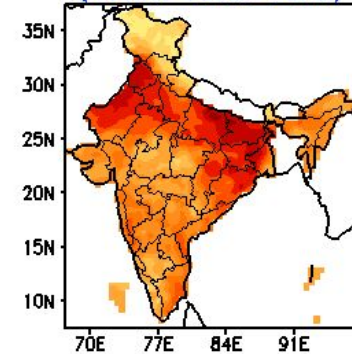
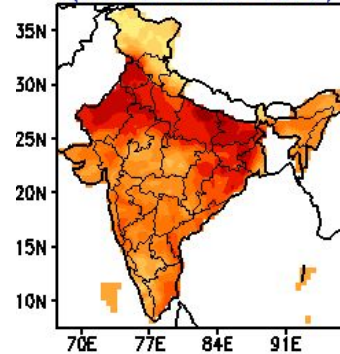
(Week1: 18Jun-24Jun)

(Week2: 25Jun-01Jul)



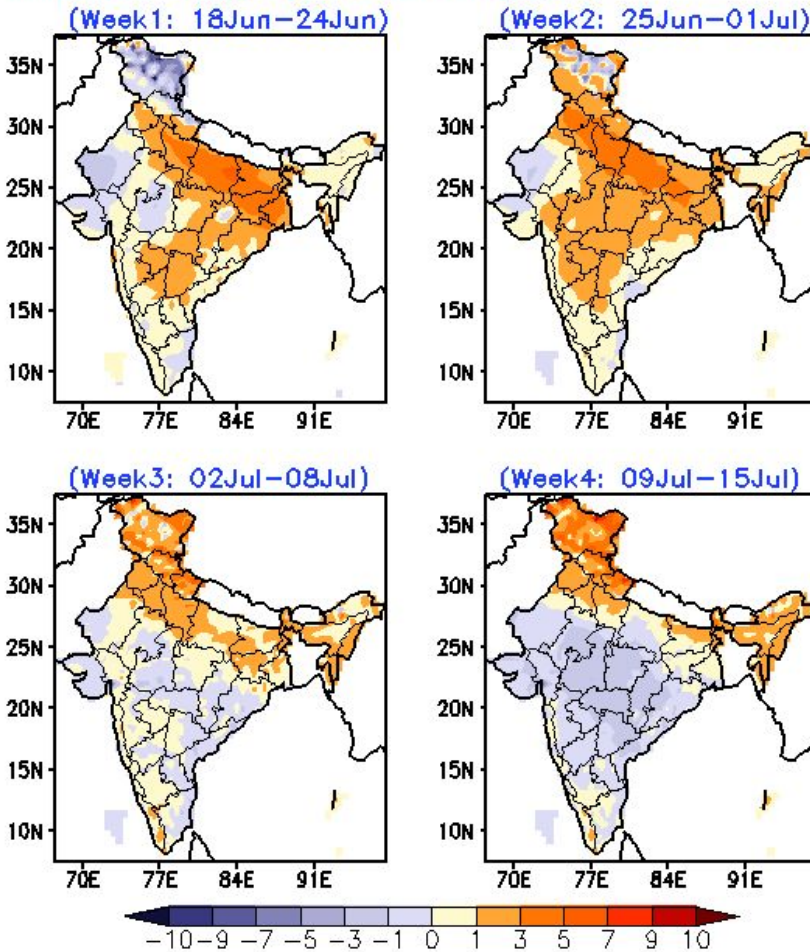
(Week3: 02Jul-08Jul)

(Week4: 09Jul-15Jul)

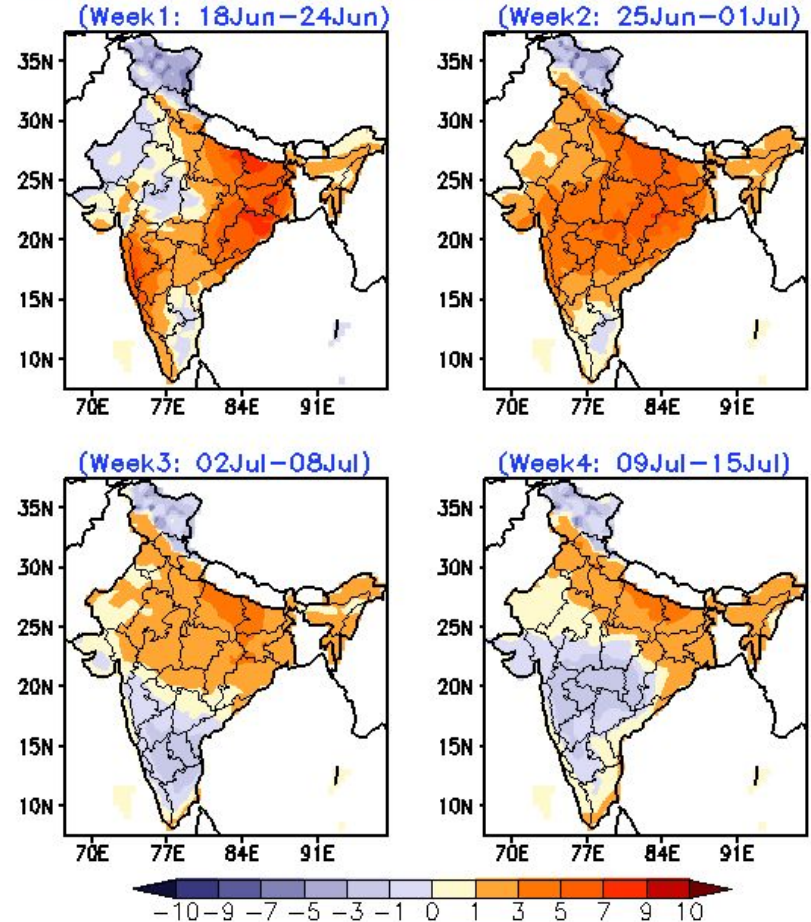


Predicted week wise temperature anomaly (by MPME)

Minimum Temperature Anomaly ($^{\circ}\text{C}$) IC=20260817



Maximum Temperature Anomaly ($^{\circ}\text{C}$) IC=20260817

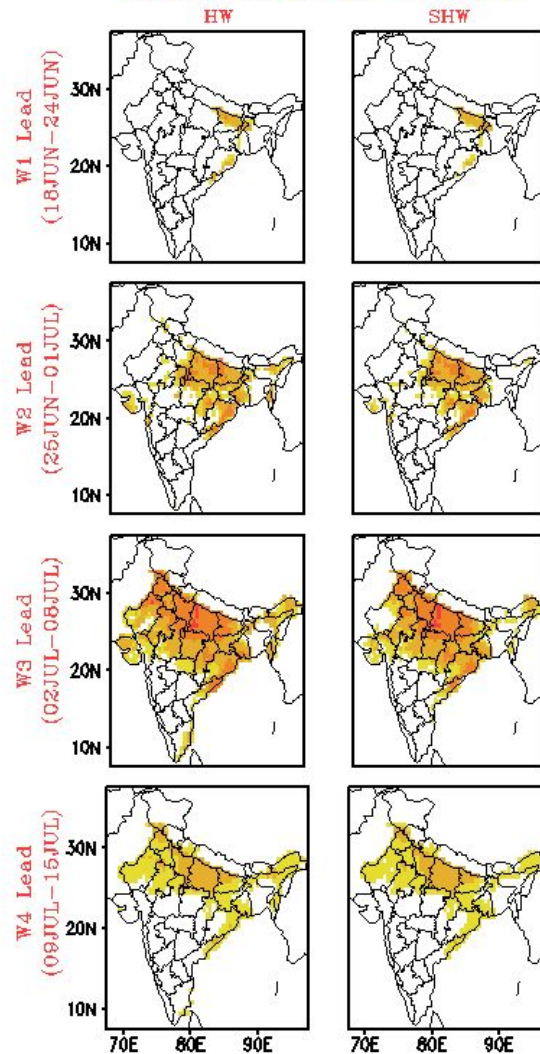


Probability of heat waves and severe heat waves

HW: Heat wave

SHW: Severe heat wave

Prediction from IC=20260617

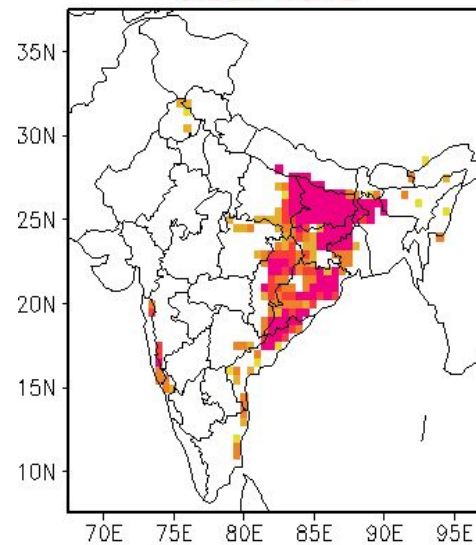


Forecast Valid Time=18JUN2026

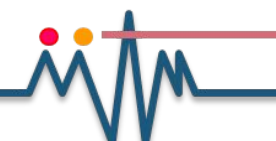
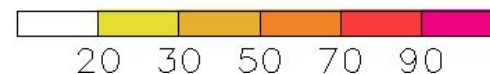
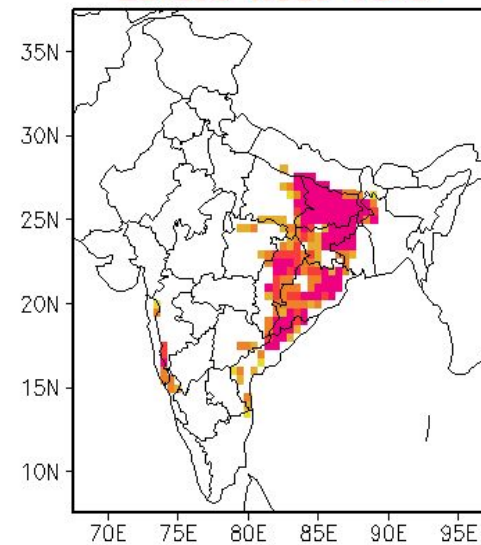
IC: 20260617

Probability of Occurrence for:

Heat Wave



Severe Heat Wave



Thanks.....

